Sarah Lang

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EDUCATION

University of Rhode Island Graduate School of Oceanography, Ph.D. candidate, Expected: June 2026 **(**GPA: 3.93)
University of Virginia, Environmental Science, B.S. with Highest Distinction**,** *College Science Scholar* (GPA: 3.83) 2021
DIS Study Abroad in Copenhagen, Icelandic Field Study, Fall 2019

RESEARCH AND WORK EXPERIENCE

Graduate Researcher, Ocean Carbon, Imaging, and Autonomous Platforms Lab, University of Rhode Island, Fall 2021 – Present, Advisor: Dr. Melissa Omand

- Analyzed data from airborne sensors (hyperspectral ocean color, surface currents from NASA DopplerScatt and SIO DoppVis), satellites (e.g. Sentinel-3), ship-based flow-through instruments, and ship-based profilers to understand the impacts of submesoscale dynamics on phytoplankton variability and carbon export in the California Current System during NASA's Submesoscale Ocean Dynamics Experiment.
- Merging phytoplankton community variability captured by NASA PACE (hyperspectral satellite), novel AVISO products (NeurOST surface geostrophic currents and SWOT wide swath altimetry), and Lagrangian particle tracking methods (OceanParcels) to characterize the impacts of eddy trapping, stirring, and dispersal on the evolution of phytoplankton ecosystems within and surrounding a cyclonic Gulf Stream ring.

Graduate Student Mentor, Student Airborne Research Program (SARP), National Aeronautics and Space Administration (NASA), Summer 2025, Faculty Mentor: Dr. Tom Bell (WHOI)

- Advised six undergraduate students conducting independent research projects on a wide range of topics within ocean sciences. Taught students how to work with a variety of satellites and large datasets in Python and JupyterLab.
- Helped students prepare AGU-style presentations. Mentorship and teaching were well received, as communicated through my final performance review and the caliber of student's final projects.
- Participated and facilitated learning during the SARP airborne campaign onboard the P-3 aircraft. Helped lead field work at the UVA Coastal Research Center collecting drone imagery and conducting quadrat surveys for multispectral imaging validation.

Independent Researcher, Castorani Lab and Scott Doney Computational Biogeochemistry Lab, University of Virginia (UVA), Summer 2019 – Spring 2021, Advisors: Dr. Max Castorani and Dr. Scott Doney

- Statistically modeled coastal water clarity in an optically complex coastal water body (Virginia Coast Reserve) vulnerable to climate change. Worked with generalized additive models, Levenberg-Marquardt algorithm, etc. in conjunction with satellite algorithms. Processed Level 1 satellite imagery with NASA SeaDAS. Published a first-author paper in AGU's Earth and Space Science and an honors thesis.

Undergraduate Researcher, Ice and Ocean Group, **UVA**, Spring 2020, Advisor: Dr. Lauren Simkins

- Used trace elemental analysis, X-Ray fluorescence (XRF), and principal component analysis to determine sediment source from samples taken from the Ross Sea in Antarctica.

NSF Research Experience for Undergraduates (REU), UVA, Summer 2019, Advisors: Dr. Max Castorani and Dr. Scott Doney

- Created a statistical satellite model for estimating Secchi depths in a coastal ocean lagoon system off the Eastern Shore of VA using R, MATLAB, and NASA SeaDAS. Analyzed seasonal and temporal changes in water clarity.

PUBLICATIONS

- **S. E. Lang**, M. M. Omand, L. Lenain. 2025. Airborne remote sensing of concurrent sub-mesoscale dynamics and phytoplankton. *Earth and Space Science.* https://doi.org/10.1029/2025EA004285
- **S. E. Lang**, K. M.A. Luis, S.C. Doney, O. Cronin-Golomb, M. CN. Castorani. 2023. Modeling Coastal Water Clarity Using Landsat-8 and Sentinel-2. *Earth and Space Science.* https://doi.org/10.1029/2022EA002579
- **S. E. Lang**, M. M. Omand, M. A. Freilich, A. Y. Shcherbina, J. T. Farrar, L. Middleton, E. Rodríguez. Estimating submesoscale-driven particulate organic carbon flux from remote sensing at a dynamic upwelling front. In revision at JGR: Oceans. Preprint available at 10.22541/essoar.176366012.20405879/v1
- J.T. Farrar, et al. S-MODE: the Sub-Mesoscale Ocean Dynamics Experiment. 2025. *Bulletin of the American Meteorological Society* https://doi.org/10.1175/BAMS-D-23-0178.1
- S. Gangrade, M.A. Freilich, É. Beaudin, K. M.A. Luis, I. Leyba, M. Omand, **S. E. Lang,** L. O'Neill, D. Capone. Open ocean biogeochemical impacts of extreme terrestrial precipitation. 2026. Submitted
- **S. E. Lang**, A.E. Jones-Kellett, M. Cornec, T. Snow, M. Masud-Ul-Alam, M. M. Omand. Lagrangian evolution of phytoplankton patchiness and communities in a cyclonic Gulf Stream eddy. In prep
- É. Beaudin, D. A. Capone, **S. E. Lang**, M. Omand, E. DiLorenzo, M. Décima, M. Freilich, Submesoscale Eddy Trapping Supports Planktonic Activity Beyond the Coastal Upwelling Zone. Submitted

PUBLISHED DATASETS AND REPORTS

- S. Lang, M. Omand, L. Lenain (2025). Data for: Airborne remote sensing of concurrent submesoscale dynamics and phytoplankton. UC San Diego Library Digital Collections. https://doi.org/10.6075/J0RN386Z
- **S. Lang**, et al., 2023. S-MODE Shipboard Flow-through Bio-Optical Measurements Version 1. Ver. 1. PO.DAAC. https://doi.org/10.5067/SMODE-RVBO2
- **S. Lang**, et al., 2022. S-MODE Shipboard Bottle Data Version 1. Ver. 1. PO.DAAC. https://doi.org/10.5067/SMODE-RVBOT
- **S. Lang,** M. Omand, & R. P. Kelly. 2022 S-MODE Bio-optical Data (V2) Report for IOP-1.
- **S. Lang,** M. Omand, & R. P. Kelly. 2023 S-MODE Bio-optical Data (V2) Report for IOP-2.
- **S. Lang,** M. Omand, & R. P. Kelly. 2022 S-MODE Bio-optical Data (V2) Report for PFC.
- **S. Lang**, O. Cronin-Golomb, and M.C.N. Castorani. 2022. Satellite-based remote sensing of water clarity Environmental Data Initiative. https://doi.org/10.6073/pasta/fe66683665a0133b2d831e552ecbae10

FIELD WORK (total days at sea: 60)

NASA Sub-Mesoscale Ocean Dynamics Experiment (S-MODE) Intensive Operations Period 2 (IOP-2) Research Cruise, April 8 – May 3, 2023

- Set-up and operated the optical flow-through system, including an ac-s Spectral Absorption and Attenuation Sensor (Sea-bird Scientific), Imaging Flow-Cytobot (McLane Labs), chlorophyll-fluorometer (Sea-bird), and C-Star transmissometer (Sea-bird).
- Calibrated ship-based and autonomous-vehicle based (e.g. Saildrones, Wavegliders, Seagliders) optical instruments for the 3 S-MODE field campaigns. Created optical proxies for chlorophyll-a and particulate organic carbon.
- Sampled and filtered seawater samples for chlorophyll-a, nutrients, pigments, carbon, etc.
- Contributed to the planning of bio-optical sampling strategies.

NASA Sub-Mesoscale Ocean Dynamics Experiment (S-MODE) Intensive Operations Period 1 (IOP-1) Research Cruise, October 3 – November 1, 2022

- Similar responsibilities to S-MODE IOP-2.

University of Virginia's Anheuser-Busch Research Center at the Virginia Coast Reserve coastal water quality and ecological sampling, Summer 2019

SELECTED CONFERENCES AND PRESENTATIONS

- **S. Lang**, et al., Estimating submesoscale-driven particulate organic carbon flux from remote sensing. Oral. Ocean Sciences Meeting 2026, Glasgow, Scotland
- **S. Lang**, et al., Estimating submesoscale-driven particulate organic carbon flux from remote sensing. Oral. S-MODE 2025 Science Team Meeting at NASA Ames, May 2025, Mountain View, CA.
- **S. Lang**, et al., Airborne remote sensing of phytoplankton and physics at the sub-mesoscale with SIO MASS. Oral. S-MODE 2024 Science Team Meeting at NASA Ames, October 2024, Mountain View, CA.
- **S. Lang**, et al., Airborne remote sensing of phytoplankton and physics at the sub-mesoscale. Oral. The University of Washington's Physical Oceanography Seminar, October 2024, Seattle, WA.
- **S. Lang**, et al., Airborne relationships between fine-scale dynamics and ocean color. Poster. OCB Summer Workshop, June 2024, Woods Hole, MA.
- **S. Lang**, et al., Submesoscale dynamics and biological distributions from airborne remote sensing. Poster. Ocean Sciences Meeting, March 2023, New Orleans, LA.
- **S. Lang**, et al., Ship-based and autonomous bio-optical measurements and calibrations. Poster. S-MODE 2023 Science Team Meeting at NASA Ames, August 2023, Mountain View, CA.
- **S. Lang**, et al., S-MODE (IOP-2). Poster. OCB Summer Workshop, June 2023, Woods Hole, MA.
- **S. Lang**, M. Omand. Biological Observations from S-MODE Poster. OCB Workshop, June 2022, Woods Hole, MA.
- **S. Lang**, K. Luis, S. Doney, O. Cronin-Golomb, M. Castorani. Modeling Coastal Water Clarity Using Landsat-8 and Sentinel-2. Poster. Ocean Sciences Meeting, March 2022, Online.
- **S. Lang**. Modeling Coastal Water Clarity Using Landsat-8 and Sentinel-2. Anchor Talk. Coastal and Estuarine Research Federation 26th Biennial Conference, November 2021, Online.

COMPUTER AND TECHNICAL SKILLS

Python, MATLAB, Cloud computing (CryoCloud) high performance computing (Unity Cluster), OceanParcels, NASA SeaDAS (GUI and CLI), REMSEM ACOLITE (GUI), R, GIS, Arduino and soldering, ocean modeling with Oceananigans (proficient)

HONORS AND AWARDS

- NASA Group Achievement Award for the Sub-Mesoscale Ocean Dynamics Experiment (S-MODE) "For Outstanding Achievements in advancing the understanding of small-scale ocean dynamics and their role in the Earth's Climate System," Mountain View, CA 2024
- Webb Family Graduate Fellowship in Oceanography, URI Graduate School of Oceanography **$3,715**
- NASA-Rhode Island Space Grant Graduate Student Fellowship, Providence, RI 2022 **$26,138**
- Wilbur A. Nelson Award, University of Virginia, Charlottesville, VA 2021 **$5,000**
- Virginia Space Grant Consortium Undergraduate Research Scholarship, Hampton, VA 2020 $**8,500**
- Patrick Sean Murphy '87 Scholarship, University of Virginia, Charlottesville, VA 2020 **$8,000**
- Raven Society, University of Virginia, Charlottesville, VA 2020

MEDIA AND OUTREACH

- Featured in *NASA* blog: PACE Hosts First Hackathon by Sara Blumberg
- Panelist, NASA's Earth Day event at the Chabot Space and Science Center for K-12 students: "Dr. Brenna Biggs Presents "Wave" Hello to NASA S-MODE: A Study of Sub-Mesoscale Ocean Processes"
- **S. Lang.** *NASA blogs: NASA's S-MODE Mission: "Sea-ing" through Rainbow-Colored Glasses, using ocean color to illuminate the biogeochemistry of the ocean*
- **S. Lang.** *NASA blogs: Life at Sea: A "First-Timer" Chronicles NASA's S-MODE Field Campaign* by Sarah Lang
- Featured in a NASA Goddard video: *A Month at Sea: Scientists Prepare to Set Sail for NASA's S-MODE Mission* at https://www.youtube.com/watch?v=aaJoF5X6a24
- *The new wave of oceanographers: Sarah Lang. PhD student focusing on how submesoscale ocean dynamics affect phytoplankton communities and plankton distribution.* SWOT AdAC Consortium Blogs

REMOTE SENSING TRAINING AND WORKSHOPS

- NASA PACE Hackweek, August 2024, Baltimore, MD
- What's behind the curtain of the NASA Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) mission? (NASA, Goddard, University of Baltimore Maryland, SRON/AIRBUS NL), August 2022, Baltimore, MD
- Calibration & Validation for Ocean Color Remote Sensing a.k.a. the Ocean Optics Class (4-week intensive workshop sponsored by NASA, University of Maine, and Bowdoin College), July/August 2021, Orr's Island, ME
- NASA ARSET trainings: Fundamentals of Remote Sensing; Hyperspectral Data for Land and Coastal Systems

EXTRACURRICULAR ACTIVITIES

2026-present Lagrangian Biophysics Working Group
2024-2026 UW Sub-mesoscale Journal Club
2024-2025 S-MODE Sub-mesoscale Journal Discussion Club

2023-2024 Educational Policy Committee, Student Representative (URI)
2023-2024 Physical oceanography departmental seminars, Co-coordinator (URI)
2023 Research advisor / mentor for North Smithfield High School senior project
2022-2023 URI-GSO graduate mentorship program, Mentor (URI)
2022-2023 Unlearning Racism in Geosciences, Member (URI)
2018-2019 Madison House Youth Mentoring, Program Director (UVA)
2018-2019 Write Climate, Vice President (UVA)